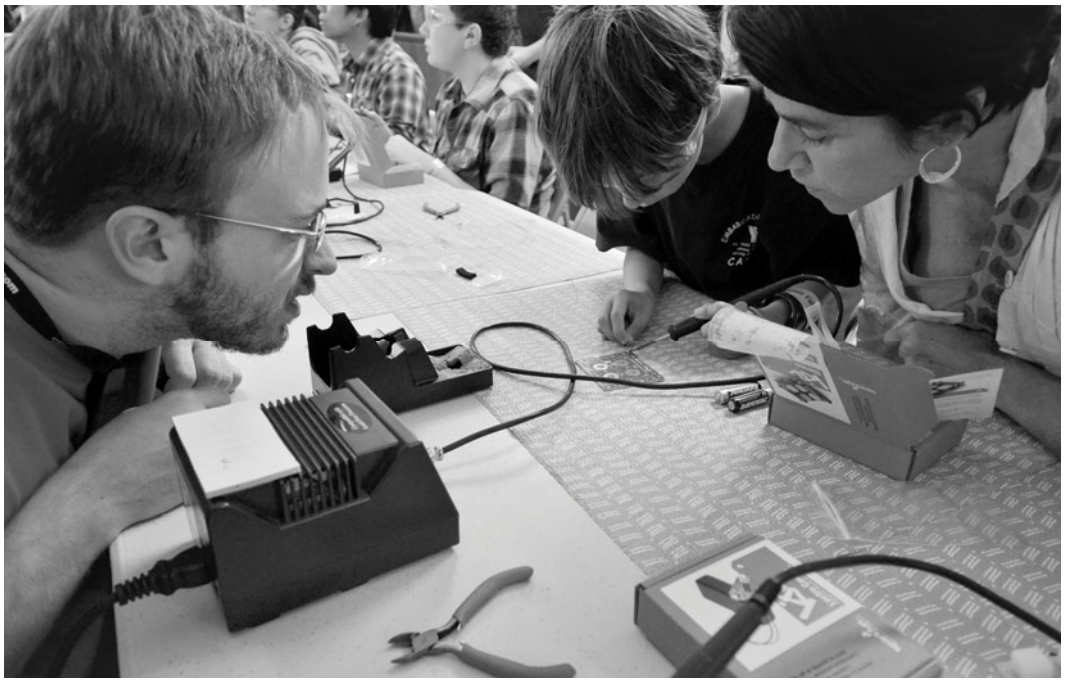


Figure 1-4. *Soldering Workshop, Maker Faire, San Mateo, CA, 2011, courtesy SparkFun Electronics*

Community engagement for a project like the Arduino is essential for the success or failure of the project. Fortuitous timing also helped in that the Arduino came to the scene at the same time that the return of the Maker and DIY movements also began again. This ever-increasing new crop of makers means that you never have to look hard to find willing co-conspirators to help on anything open source; whether it's through pockets of helpful individuals in the interweb or your neighborhood hacker space, help is never that far away.

Arduinoland

This community-driven research and development seems to have a magical effect on any project to come out of the wake of the Arduino, and this proverbial kingdom of mystical wonder is something I'm going to call Arduinoland. Maybe it's because the hardware design works, is open, and is hackable, or maybe it's because of a communal desire to build upon each other's work, but whatever it is, in Arduinoland, soldering iron marvels, difficult to nearly impossible to pull off in a garage just ten years ago, seem to happen every day. Anything from interactive electronic textiles, autonomous flying aircraft, networked beer keg refrigerators, photographic missions to near space, and immersive architectural installations have all sprouted from this open Arduino ecosystem.

Take the MakerBot CupCake CNC for example, a sub-$1,000, kit-based, plastic-extruding, 3D printer shown in Figure 1-5. Needing an electronics system to interpret codes sent from an attached computer to drive the positioning of the printhead and the temperature of the extruder, the MakerBot team built off the Arduino hardware platform and the development environment to create an open platform for building plastic objects.